Received 11 December 2024; revised 5 February 2025; accepted 13 February 2025. Date of publication 17 February 2025;
date of current version 13 March 2025. The review of this article was arranged by Editor S. Reggiani.

Digital Object Identifier 10.1109/JEDS.2025.3543017

Harmonic Enhancement of Terahertz GaN Planar Gunn Oscillators With Multiple Gates

YING WANG, SHUAI HUI, YU-XIN FU, YUAN-ZHU XIA, AND HE GUAN

School of Microelectronics, Northwestern Polytechnical University, Xi'an 710072, China

CORRESPONDING AUTHOR: H. GUAN (e-mail: he.guan@nwpu.edu.cn)

This work was supported in part by the National Natural Science Foundation of China under Grant 62104199; in part by the Natural Science Basic Research Program of Shaanxi for Youth Projects under Grant D5110210132; in part by the National Natural Science Foundation of China under Grant 62471402; in part by the National Key Research and Development Plan of China under Grant 2023YFB2905200; and in part by the Key Research and Development Plan of Shaanxi Province, China, under Grant 2023GXLH-084.

ABSTRACT In this paper, we propose a novel design of GaN-based planar Gunn oscillator as terahertz signal source. The oscillator has multiple gates and each gate can be individually biased. By controlling gate bias voltage and distance to the barrier layer, the output oscillating current can be formed in such a way that the higher harmonics are more powerful than the fundamental one. This is because multiple Gunn domains created under the gates in the channel are synchronized. Compared to the conventional single-domain diode, this multi-gate configuration not only increases the frequency and output power but also provides superior harmonic control, leading to higher efficiency and more stable operation at terahertz frequencies. We will review the design details and analysis on domain forming conditions using a physics-based numerical model. The optimal parameters for high power, high DC-RF conversion efficiency and high frequency will be given. Specifically, in dual-gate device, when it works in the dual-domain mode, the second harmonic is enhanced, reaching a frequency of 310.5 GHz with 7.6 mW of power and 11.2% efficiency. The tri-gate device operating in tri-domain mode further enhances the third harmonic to 417.0 GHz, with 9.57 mW of power and 9.23% efficiency. The multi-gate structure allows for more efficient harmonic generation, greater frequency tunability, and better power management, all of which are crucial for advanced terahertz application such as mixing, frequency multiplexing, and signal amplification.

INDEX TERMS III-V semiconductors, terahertz devices, Gunn effect oscillator, harmonic enhancement.

I. INTRODUCTION

Terahertz (THz), located between infrared and microwave on the electromagnetic spectrum, has many advantages such as wide bandwidth, low phonon energy, "see through" capability, fingerprint characteristics, security etc [1]. Typical applications are found in the fields of space exploration, wireless communication, security screening, material science, biomedical engineering and so on. However, the development and application of THz science and technology are largely limited due to lack of high-performance terahertz sources and detectors [2].

Solid-state sources have distinct advantages such as compact size, easy integration, low cost, and mass production benefiting from the advancements of semiconductor device manufacturing industry in the past decades [3]. Gunn diodes, as one of the important solid-state oscillators, have attracted increasing attention and been widely deployed in various fields since they were reported in 1963 [4], [5]. Recent years, GaN hetero-structural planar Gunn diodes have emerged and become one of the most promising THz oscillators for the potentials of high frequency, high power and high temperature [6], [7]. In 2008, Hajo et al., experimentally verified the Gunn effect in GaN diodes for the first time by observing bias oscillations using series inductors [8]. At the same time, Xu et all, simulated the InGaAs self-switching nanodiode (SSD) using a Monte Carlo model, which generated an oscillation frequency of 130 GHz and a peak current of approximately 10 μA [9]. By 2010,

© 2025 The Authors. This work is licensed under a Creative Commons Attribution 4.0 License.
For more information, see https://creativecommons.org/licenses/by/4.0/

Hajo et al., used side contact technology to achieve stable operation of up to 22 volts on a Gunn device based on a 2 μm thick GaN with a device current of about 1 A, resulting in high output power [10]. In 2020, Hajo et al., experimentally verified the existence of high NDR in Gan-based Gunn diodes by using side contact and field plate techniques. The GaN Gunn diode they designed has an effective region of about 600 nm and is expected to operate at frequencies up to 330 GHz [10]. In 2021, Lee et al., demonstrated a defection-free GaN diode using a Monte Carlo-A model capable of generating sustained current oscillations up to 625 GHz for Gunn diodes with transmission lengths as short as 500 nm [11]. In 2023, Hajo et al., used a magnetic field plate and side contact technology to create a diode that exhibits a high NDR, a current drop of about 100 mA, and an expected fundamental frequency of 330 GHz [12]. In 2024, García-Sánchez et al., Pointed out that the lack of successful GaN Gunn diodes is attributed to impact ionization caused by buffer leakage current and high electric fields at the anode corner, when biased above 20 V. By means of Monte Carlo, they proposed that leakage current can be effectively suppressed by Schottky substrate terminal, and vertical cathode-substrate current can be avoided by use a MIS configuration for the substrate contact [13]. Due to the limitations of AlGaN material, GaN material growth process and HEMT device manufacturing process, the manufacture of GaN planar Gunn devices faces certain challenges, and there are still many theoretical studies in this field. Compared to traditional vertical Gunn diodes in which electrical currents pass through epitaxial materials, the planar Gunn diodes in which electrical currents pass in parallel to epitaxial layers have striking advantages such as easy integration, simple structure, and controllable frequency. In addition, the planar design avoids the issues of high junction temperature and large package volume that traditional vertical Gunn devices encounter, and the planar Gunn diodes are more suitable for Monolithic Microwave and Terahertz Integrated Circuit (MMIC, MTIC) [14], [15]. GaAs and InGaAs-based planar Gunn diodes have been thoroughly studied elsewhere [16] and the reported results show limitation of output power and oscillation frequencies. However, theoretical studies on GaN-based Gunn diodes showed that the output power could be as high as several hundred milliwatts in terahertz band [11], [17] and the oscillating frequency could reach as high as 2.0 THz [18].

In theory, the formation and transportation of electronic field domains within Gunn devices determine the output characteristics such as frequency and power. By manipulating the domains, the performance of the Gunn device can be optimized effectively. Single domain has widely been investigated in the literature [19]. According to the single domain theory, it is difficult to improve the oscillation frequency and output power at the same time. This is because based on the single-domain theory, the oscillation frequency of the device can only be increased by reducing the channel length. For example, based on some theoretical work [11] to achieve the operating frequency of 326 GHz, the device's transit length is only 700 nm. From the experimental results, based on the traditional single domain model, in 2012, Khalid et al. fabricated the first $In_{0.53}Ga_{0.47}As$/InP planar Gunn diode oscillating at fundamental modes above 160 GHz with the maximum output power is 98 μW [20]. In 2014, Khalid et al. fabricated the 600-nm-Gunn diode with the operating frequency of over 300 GHz, which is the highest experimental oscillation frequency so far [21]. Implementing such short-channel single-domain Gunn devices is not only technologically complex, but also yields a mere 28 μW of RF output power. As the device length is shortened, on the one hand, the size of the electronic domain will decrease, leading to a decrease in the device's output power; on the other hand, the difficulty of device manufacturing will significantly increase, making it difficult to achieve good ohmic contact, and the self-heating effect of the device will also aggravate. In our previous published article, we have proved the multi-domain phenomenon in Gunn devices. The extremely high concentration of 2DEG in the heterogeneous junction of GaN easily leads to the complex multi-domain phenomenon. Same or different types of electric field domains are formed simultaneously during one oscillation circle [22], [23], [24]. In 2016, Ai min Song et al., also observed multi-domain properties based on planar Gunn diodes with quantum well structure of AlGaAs/InGaAs, using "trapezidal" cathode contacts to simulate non-uniform metal contacts caused by etching manufacturing [25]. Without appropriate control on multi-domains, the device would exhibit poor noise performance because the unsynchronized multi-domains generate multiple frequency tones that suppress the fundamental one. On the contrary, purposefully manipulated stable and synchronized multi-domains will not only enhance the harmonic components, but also lead to new applications such as frequency multiplication or mixing. Although multi-domain phenomena have been observed in some experiments [26] there lacks of theoretical investigations. Nevertheless, the phenomenon of multi-domain undoubtedly provides new opportunities for device engineering.

In this paper, we take advantage of the multi-domain phenomenon and propose a novel GaN-based planar Gunn diode which has meticulously designed multiple gates. By controlling gate bias voltage and distance to the barrier layer, the output oscillating current can be formed in such a way that the higher harmonics are more powerful than the fundamental one. Therefore, synchronous improvement of device frequency and power has been achieved. The gates are separated vertically and horizontally by Si_3N_4 in a stagger way between anode and cathode. By optimizing metal work-function, gate length and the distance between the gate and the barrier layer, the number of nucleating points of the domains in the channel can be controlled. Furthermore, the formation sites of domains can be regulated by adjusting the biasing voltage on the gates. Thus, the

output oscillation frequency of the diode can be electronically tuned to any harmonic components. For example, in dual-gate devices, electronic domains can be formed independently or simultaneously at either gate by adjusting the gate voltage. If the device operates in tri-domain mode, the third harmonic component is greatly enhanced, with f=417.0 GHz, P_{RF}=9.57 mW, and η=9.23% to achieve a simultaneous increase in frequency and power. The output characteristics significantly exceed the international highest standards [20], [21], which proves that it is an effective way to improve the frequency and power simultaneously by the modulation of multi-domain mode to enhance the harmonic component. Simultaneously, almost all the terahertz devices based on Gunn effect, whether they are oscillators, mixers or amplifiers, face the multi-domain phenomenon. Therefore, the traditional single-domain theory is not adequate for the research and design of Gunn devices. The multi-gate Gunn diode can freely modulate the number of the electron domains, which makes it easier to study of complex domain modes. Here is the outline of this paper. In Section II, we will explain the design of the proposed multi-gate oscillator and describe the simulation method. In Section III, we will present an explicit analysis on the physical mechanism and output characteristics of this novel diode. Finally, a conclusion is given in Section IV.

II. DEVICE DESIGN AND PHYSICAL MODEL

Figure 1 depicts a cross-sectional schematic of the proposed GaN planar Gunn diode. The epitaxial structure of the device is composed of a 22 nm $Al_{0.3}Ga_{0.7}N$ barrier layer, a 20 nm GaN channel layer, a 25 nm $Al_xGa_{1-x}N$ (x: 0.1-0) back barrier layer and a 2 μm GaN buffer layer. All layers are unintentionally n-type doped and have a background carrier concentration of 1×10^{15} cm^{-3}. The AlGaN back barrier layer is adopted to enhance the confinement of heterojunction two-dimensional electron gas (2DEG) and prevent electrons from spilling into the buffer layer which results in reduction of mobility and reliability. Some reports argue that inappropriate back barrier layer design can potentially introduce a parasitic channel at the AlGaN back barrier/GaN buffer interface [27]. In this study, we choose to use a graded AlGaN back barrier with Al mole concentration changing linearly from 0.1-0. This design has been approved to be able to weaken polarization between AlGaN/GaN buffer and avoid creation of parasitic channels.

Cathode and anode are ohmic contacts and deliberately placed on the top of the back barrier to avoid non-converging problem of the model. We set the total effective channel length 1.2 μm. Three gates are buried in Si_3N_4 and have equidistant from anode as shown in Fig. 1. The first gate is not attached to cathode to ensure high-quality ohmic contact. The distance between the first gate and cathode is set to be L_1=50 nm in all the simulations. The purpose of introducing the gates is to control the formation of electric field domains in the channel. We set all the gate lengths L_g to be 100 nm in this work. All gates are set to be equidistant, with a spacing

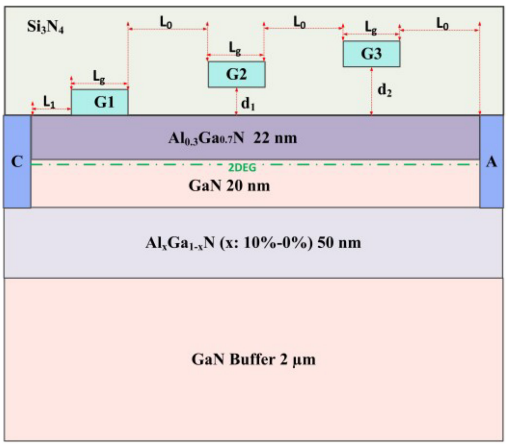


FIGURE 1. Cross-sectional schematic diagram of the proposed AlGaN/GaN tri-gate tunable Gunn diode.

of l_0. Except the first gate which is directly deposited onto the top AlGaN barrier layer, the other two gates are fully merged in the Si_3N_4 insulation layer. The distance of the Si_3N_4 between gate and AlGaN barrier layer is designated as d_i (i= 1, 2). The relative permittivity of Si_3N_4 is 7.0. The device width is 1 μm and all simulations are carried out at 300 K.

Other physical models such as Fermi-Dirac, low field and high field electron mobility, Shockley-Read-Hall (SRH) recombination and polarization models are adopted in this simulation. The model used in this two-dimensional numerical simulation is the more advanced energy balance model instead of the drift-diffusion approximation. This is because the drift-diffusion approximation is less accurate for smaller feature sizes and high operation oscillation. Energy balance mode contains the non-local carrier transport effects, such as velocity overshoot and non-local impact ionization, which are not included in the drift-diffusion model. The energy relaxation time, t_e and momentum relaxation time, t_m of GaN are set to be 500 fs and 4 fs based on [28], [29], [30].

Polarization model is critical for GaN hetero-structural devices. The total polarization (P) is the sum of spontaneous Psp and piezoelectric polarizations P_{pe}. The spontaneous and piezoelectric polarization effects can be estimated by the following equations (2-1) and (2-2) [18]:

$$P_{sp}(x) = xP_{sp}(AlN) + (1-x)P_{sp}(GaN) \quad (2.1)$$

$$P_{pe}(x) = 2\frac{a_{GaN}-a(x)}{a(x)}\{e_{31}\quad(x) - e_{33}(x)\frac{C_{13}(x)}{C_{33}(x)}\} \quad (2.2)$$

$$a(x) = xa_{AlN} + (1-x)a_{GaN} \quad (2.3)$$

where, a (x) and a_{GaN} are lattice parameter of $Al_xGa_{1-x}N$ and GaN, respectively. The graded back barrier layer is defined as complete lattice relaxation. Therefore, we don't take the piezoelectric polarization into consideration. Thus, the total polarization of the back barrier layer is equal to spontaneous polarization (P_{sp}). If the Al composition of

TABLE 1. Parameters of GaN, AlN, and Si_3N_4 in the simulation [34], [35], [36].

Parameter	GaN	AlN	Si_3N_4
Band Gap (Eg/eV)	3.42	6.25	5.0
Permittivity (ε)	9.0	8.5	7.0
Hole mobility (cm2/Vs)	22	14	
Spontaneous polarization P_{sp} (C/m^2)	-0.029	-0.081	
Lattice parameter a (Å)	3.189	3.122	
piezoelectric coefficient e_{33} (C/m^2)	1.27	1.79	
piezoelectric coefficient e_{31} (C/m^2)	-0.35	-0.50	
elastic constant C_{13} (GPa)	106	108	
elastic constant C_{33} (GPa)	398	373	

AlGaN back barrier layer varies along the growth direction Z, the corresponding density (cm^{-3}) is given by:

$$\rho_P = -\nabla \cdot P = -\frac{\partial P}{\partial Z} \tag{2.4}$$

To calculate the output RF characteristics of the diode, a sinusoidal voltage in the form of $V_{dc}+V_{ac}\sin(2\pi ft)$ is added across the anode and cathode of the diode to reduce the calculation time. More details regarding this setup can be found in [29]. The efficiency of the power conversion is defined as $\eta=-P_{RF}/P_{dc}$ (P_{RF} is output oscillation power; P_{dc} is the DC power). The power calculation methods of P_{RF} and P_{dc} are given below [31], [32], [33].

$$P_{RF} = \frac{V_{ac}}{T_R}\int_0^{T_R} I(t)\sin(2\pi ft)dt \tag{2.5}$$

$$P_{dc} = \frac{V_{dc}}{T_R}\int_0^{T_R} i(t)dt \tag{2.6}$$

All other important physical parameters are listed in Table 1.

III. RESULTS AND DISCUSSION

In this paper, we prefer to use an ideal simulation method to demonstrate the potential of enhancing power frequency by harmonic enhancement and we don't consider the effects of self-heating, defects, and parasitic resistance in our simulations. Since multi-domain simulations are quite complex and often encounter convergence issues, incorporating these effects would cause unbearable computational load and convergence problems. In our previous research work, we have already explored the impact of self-heating on the DC curve. Except for the self-heating model, all other physical models in this paper are consistent with those in [37]. In Figure 2, we present a comparison of the I-V characteristic curves of different devices, including the ideal planar Gunn diode (PGD) without a gate in this manuscript, the PGD-on-Si considering thermal effect from our previous work [37], and the experimental measurement data of GaN HEMTs with similar structure from Ref. 40-42. Compared with the experimental data, although the ideal simulation results are

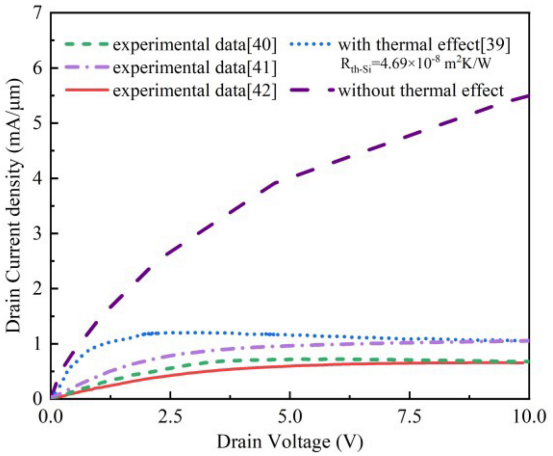


FIGURE 2. Comparison of I-V characteristics curves among the ideal GaN Gunn diode without a gate, GaN Gunn diode considering thermal effect [37], and experimental GaN HEMTs from different literatures, at Vg=0 V [38], [39], [40]. In Ref. 39, the thermal resistance R_{th} for Si substrates is defined $R_{th-Si} = 4.69 \times 10^{-8}$ m^2 K/W.

generally higher than the experimental values, the current density with thermal effect is closer to the experimental data, especially at higher voltage, which proves the credibility of our ideal simulation results.

Then we investigate GaN hetero-structural Gunn diode with single gate. For comparison, a traditional planar Gunn diode without a gate is also simulated. The channel length L of the diode without a gate is set to be 400 nm. To meet the oscillation criteria at a stable dipole domain mode, following conditions must be satisfied [12]:

$$\mathrm{N} \times \mathrm{L} \geq 5 \times 10^{13}\mathrm{cm}^{-2} \tag{3.1}$$

$$\mathrm{N} \times \mathrm{d} \geq 2 \times 10^{11}\mathrm{cm}^{-2} \tag{3.2}$$

where N is doping concentration in the channel and d is the channel width. For the epitaxial layer structure shown in Fig. 1, the average electron concentration N and the thickness of the 2DEG is approximately 5×10^{19} cm^{-3} and 4 nm, respectively. Therefore, the proposed $Al_{0.3}Ga_{0.7}N/GaN/Al_{0.1\text{-}0}Ga_{0.9\text{-}1}N$ Gunn diode should meet the criteria and operate at a dipole domain mode. However, our simulation results shown in Fig. 3(a) indicate a rather small dipole domain is formed and concentrated in a very small region near the anode and almost three quarters of channel have no domains. The part of the channel without domains, so-called a dead zone, is purely resistive and does not participate domain formation and therefore results in really small oscillating current but extremely high frequency as shown in Fig. 4. The higher frequency is due to shorter effective channel. In order to make the Gunn oscillator operate more efficiently and produce higher power, besides the requirements listed in (3-1) and (3-2), there should be adequate electric field disturbance near the cathode of the channel that accelerates and energizes the electrons to jump into upper valley in the conduction band, so as to stimulate the quick formation of dipole domains. As in all

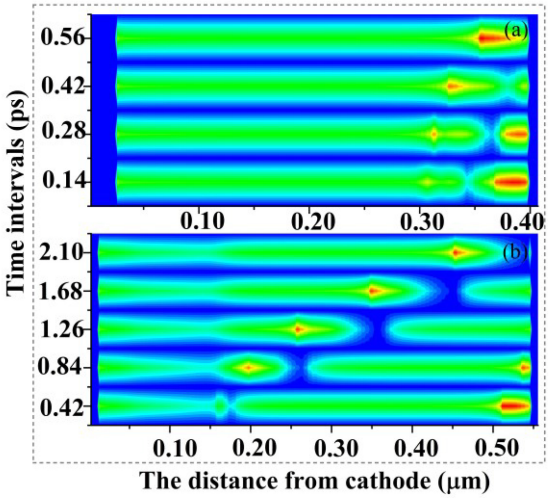


FIGURE 3. Illustration of transient formation of Gunn domains for a planar Gunn diode without a gate (a) and with a gate (b).

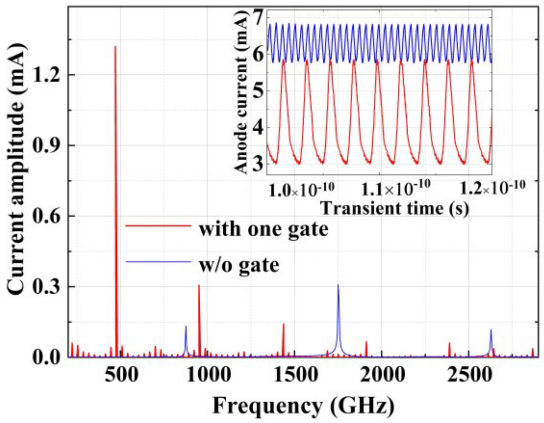


FIGURE 4. The frequency spectrum and oscillating current wave (insert figure) for The GaN planar oscillator with and without one gate.

devices we investigate here, the Ohmic anode and cathode electrodes are vertical and in good contact with the 2DEG channel, therefore no electric field disturbance forms near the cathode of the device. This configuration causes electrons to accelerate over a long distance before transferring into the upper valley and form domains. This inevitably leads to a longer dead zone and therefore lower DC-RF conversion efficiency.

In order to reduce the dead zone and promote the quick formation of the dipole domain, we propose a Schottky gate near the cathode. If the work-function (W_m) of the metal gate is greater than the work-function (W_s) of the $Al_{0.3}Ga_{0.7}N$ barrier layer, a depletion region will be formed under the gate. When W_m is high enough, the depletion region can reach the 2DEG channel where some electrons will be locally depleted. The high gradient of electron concentration between the depleted region and the channel will generate a locally enhanced electric field. The electric field peak promotes the rapid formation of electronic domains. With the increase of the metal work-function, the length of the dead zone becomes smaller and the conversion efficiency improves. When W_m reaches 4.6 eV, i.e., Molybdenum, the dead zone almost disappears completely, and the electronic domain forms almost immediately under anode edge of the gate, as shown in Fig. 3(b). The amplitude of the oscillating current is increased by almost 4.4 times comparing with the conventional diode without a gate, as shown in Fig. 4. The corresponding output power P_{RF} is about 5.25 mW, and the conversion efficiency is 15.50%. If continuing to increase W_m, the 2DEG channel will be further depleted until the channel is completely pinched off. With current design, our simulation shows the optimal W_m is 4.6 eV because that gives the minimum channel resistance. If W_m of gate metal is less than 4.6 eV, a recess into the barrier layer can be considered; on the contrary, if W_m is greater than 4.6 eV, an insulation layer can be inserted between the gate and barrier layer, thereby reducing the gate's depletion of 2DEG. Furthermore, it is well-known that the depletion depth can also be controlled by biasing voltage one the gate. We can also use this principle to control the formation and optimization of domains. Considering this, we propose multiple gates between anode and cathode of the GaN-based planar Gunn oscillator. Since metal gates can control nucleating position of Gunn domains in the channel, the fundamental oscillation frequency of and harmonics can be manipulated. Firstly, we investigate the Gunn diode with double gates. For a dual-gate Gunn oscillator, the first gate that is near the cathode is directly deposited on the AlGaN barrier layer; while the second gate that is near the anode has a thin layer of Si_3N_4 beneath. Fig. 5 depicts the change of electric field E for different thicknesses of the insulation $d_{insulation}$. As shown in Fig. 5(a), if the second gate is directly deposited on the AlGaN barrier layer (i.e., $d_{insulation}$ =0), the electric field under the second gate is much higher than that under the first gate. As a result, no electron domain can form under the first gate. The first gate and the channel between the two gates will be resistive and lossy. The electric field peak under the second gate decreases with the increase of $d_{insulation}$. As $d_{insulation}$ exceeds 40 nm, the two electric field peaks are similar in value, therefore two dipole domains form simultaneously under two gates and the second harmonics of the output oscillating current are enhanced as a result. However, if $d_{insulation}$ continues increasing the electric field will decrease and as it reaches to 120 nm,the electric field under the gate is too low to sustain the formation of a domain. Therefore, only one domain that originally comes from near the first gate remains in the channel.

With $d_{insulation}$ set to be 120 nm, we investigate how biasing voltage on gates (i.e., V_{G1} for Gate 1 and V_{G2} for Gate 2) affect the distribution of the electric field along the channel. We vary the biasing voltage of V_{G2} between −4 V and 0 V in steps of 1 V while maintain V_{G1} at 0 V. Figure 6 shows the corresponding transient oscillating current and spectrum characteristics. Figure 7 further reveals

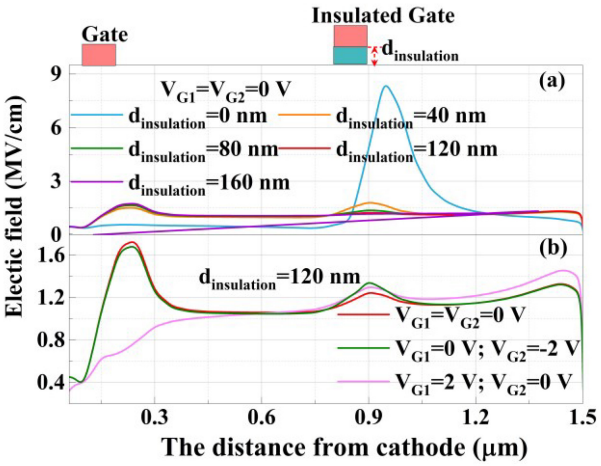


FIGURE 5. The electric field distribution curves along the channel correspond to d=0, 40, 80, 120 and 160 nm.

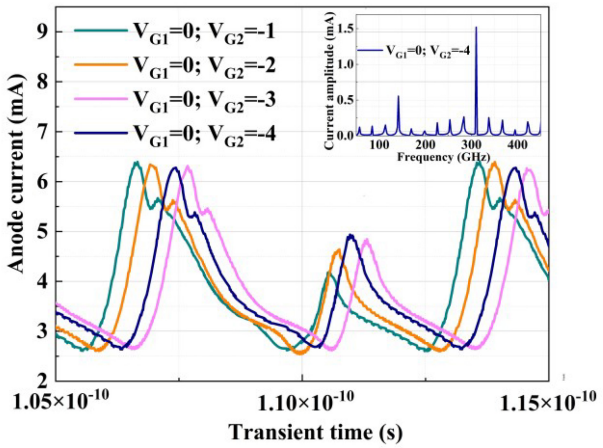


FIGURE 6. The oscillating current waves and the corresponding frequency spectrums for the DG planar Gunn oscillator at different voltage:$V_{G1}=0$ V; $V_{G2}=-1$ V; $V_{G1}=0$ V; $V_{G2}=-2$ V; $V_{G1}=0$ V; $V_{G2}=-3$ V; $V_{G1}=0$ V; $V_{G2}=-4$ V.

the two-dimensional contours of the electric field distribution along the 2DEG channel derived at the same time intervals inside one oscillating circle. Firstly, when $V_{G1}=V_{G2}=0$ V, the dual-gate oscillator only produces one fundamental frequency of 141.6 GHz with a RF power of 2.52 mW and conversion efficiency η of 6.2%. Secondly, when $V_{G1}=0$ V and the insulated gate is reversely biased, the electric field peak under the insulated gate is enhanced. Therefore, each gate forms its own dipole domain as shown in Fig. 7(c). The electron domains gradually grow up as they move towards the anode, and eventually disappear at the anode in turn. As a result of this, the waveform of the output oscillating current is no longer sinusoidal in time-domain and two peaks appear in the frequency domains as shown in Fig. 6. The smaller current peak is generated due to the disappearance of the dipole domain under the second gate, while the larger peak is generated by disappearance of the dipole domain under the first gate. Increasing the absolute value of V_{G2} will enhance the electric field under the insulated gate, which in turn increases the size of the second dipole domain and the second current peak, as shown in Fig. 6. Therefore, when $V_{G2}=-4$ V, the second harmonic is enhanced at the frequency f of 310.5 GHz, with the output power P_{RF} of 7.56 mW that leads to a DC-RF conversion efficiency η of 11.15%. Ideally, to enhance the output power and linearity of the oscillation current, the transmission time of each domain should be an integer multiple of the shortest transmission time, which requires the effective channel length of the each domain to be an integer multiple of the shortest length l_0. Therefore, for multi-gate devices, we set the gates to be equidistant. Therefore, when a dual-gate device operates in dual-domain mode, it is approximately equivalent to the superposition of two single gate devices with the effective channel lengths of l_0 and $2l_0$, thus enhancing the second harmonic component. Compared with single-gate devices, the superposition of double domains significantly increases harmonic power in dual-gate device. However, the presence of gates can also reduce the linearity of the device's oscillation current. For example, in the dual-gate device, the effective channel length for the first electron domain is $2l_0+l_g$, whereas the effective channel length for second electron domain is l_0, as shown in Fig. 1, which deteriorates the noise performance of the device. In addition, the introduction of Schottky gates introduces parasitic channel resistance [41], which may reduce the output power of the device and introduces thermal noise, etc [42], [43]. Therefore, within the process conditions, the length of the gate should be as short as possible.

Thirdly, when $V_{G2}=0$ V and the insulated gate is forward-biased, the electron concentration under the first gate will increase with the increase of V_{G1} and the electric field under the gate also decreases. As illustrated in Fig. 5, when $V_{G1}=2$ V, the electric field near the cathode and the first gate is much lower than that in other part of the channel. As shown in Fig. 7(b), only one dipole domain forms near under the insulated gate and the output oscillating current is quasi-sinusoidal (Fig. 8). The fundamental frequency that is decided by the distance between insulated gate and anode (L_0) is as high as 282.2 GHz, with a RF power of 5.20 mW and efficiency of 10.8%. This is an interesting result because by introducing a second gate and applying appropriate bias voltage, the RF power and efficiency of the second harmonic of oscillation has been significantly enhanced. Based on this, we can extend the number of gates even further, e.g., three or more to achieve more complex multi-domain coupling, and therefore the wider range of adjustable frequencies and the number of harmonics can be achieved. To demonstrate this, we have investigated tri-gate planar Gunn diodes. Similarly, except for the first gate, the other two gates are buried in the Si_3N_4 insulation layer. The closer the gate to the anode, the thicker the insulation layer is so as to make the electric field peaks value approximate each other. As shown in Fig. 1, for the proposed tri-gate Gunn oscillator, $d_1=70$ nm, $d_2=220$ nm and $L_0=400$ nm. In the

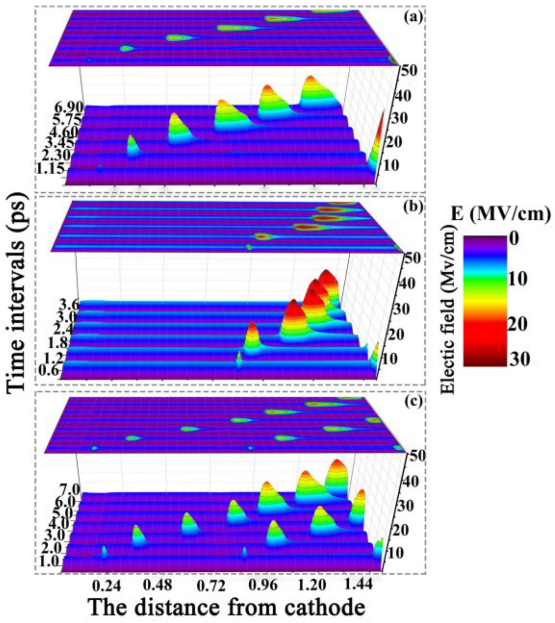


FIGURE 7. The electric field distribution curves devired at the same time intervals inside one oscillation circle for DG oscillators at the different gate voltages: (a) V_{G1}=V_{G2}=0 V; (b) V_{G1}=2 V; V_{G2}=0 V; (c) V_{G1}=0 V; V_{G2}=-4 V.

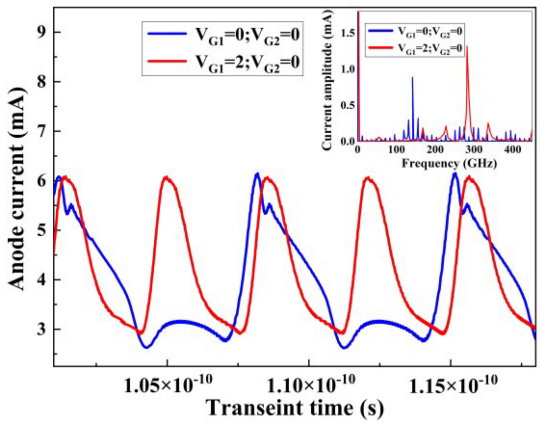


FIGURE 8. The oscillating current waves and the corresponding frequency spetrums for the DG planar Gunn oscillator at different voltage: V_{G1}=2 V; V_{G2}=0 V; V_{G1}=0 V; V_{G2}=0 V.

tri-gate oscillator, with different bias voltages on the gates, three types of independent quasi-sinusoidal oscillating waves with the fundamental frequencies of 131.5 GHz, 204.5 GHz, 460.2 GHz can be formed in the channel. However, if V_{G1} is set 0 V and the other two insulated gates are reversely biased, three dipole domains are formed under each gate, as shown in Fig. 9. The three domains exit from the anode successively, resulting in three peaks in the oscillating current as shown in the insert of Fig. 10. This configuration ensures the third harmonic component is greatly enhanced, with f=416.8 GHz, P_{RF}=9.57 mW, and η=9.23%.

Based on the analysis above, one can conclude that increasing the number of the gates could generate more faster Gunn domains that leads to enhancement of RF

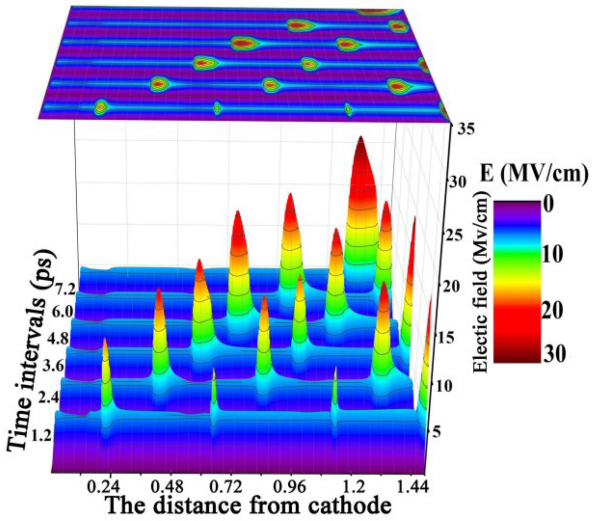


FIGURE 9. The electric field distribution curves devired at the same time intervals inside one oscillation circle for TG oscillators when V_{G1}=0 V; V_{G2}=-2 V; V_{G3}=-4 V.

TABLE 2. Comparison of the output performance of the MULTI-gate GUNN devices.

	State of gate voltages	No. of domains	Output power P_{RF}	Harmonic frequency f	Conversion efficiency η
Dual-gate diode	(V_{G1}=0V, V_{G2}=0V)	1	2.52mW	141.6 GHz	6.20%
	(V_{G1}=0V, V_{G2}=1 V)	1	5.20mW	282.2 GHz	10.80%
	(V_{G1}=0V, V_{G2}=-4 V)	2	7.56mW	310.5 GHz	11.15%
Tri-gate diode	(V_{G1}=0V, V_{G2}=V_{G3}=-4V)	3	9.57mW	416.8 GHz	9.23%

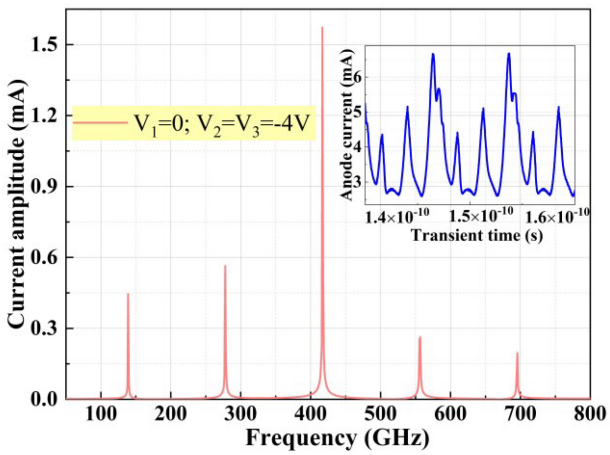


FIGURE 10. The frequency spectrum and oscillating current wave (insert figure) for the GaN tri-gate planar oscillator.

power for higher harmonic oscillations. Table 2 provides the comparison of the output performance of the multi-gate Gunn devices.

IV. CONCLUSION

In this paper, we have demonstrated a multi-gate GaN-based planar Gunn diode for the first time. Through numerical simulation, we have optimized the structure of the multi-gate diode, and revealed its work mechanism in detail. Firstly, in the multi-gate diode, we can conveniently adjust the output frequency and harmonic by controlling the number and voltage of gates. By increasing the number of gates, the adjustable range of frequency and harmonic is expanded. When V_{G1}=0V, V_{G2}=−4V, the second harmonic component of dual-gate Gunn device is greatly enhanced, with f=310.5 GHz, P_{RF}=7.56 mW, and η=11.15%. When V_{G1}=0V, V_{G2}=V_{G3}=−4V, the third harmonic component of the tri-gate Gunn device is greatly enhanced, with f=417.0 GHz, P_{RF}=9.57 mW, and η=9.23%. This work may trigger further studies on the multi-domain modulation and coupling mechanism based on multi-gate structure, which can add credits to the development of Gunn domain theory and new developments of terahertz devices.

REFERENCES

[1] S. Ergün and S. Sönmez, "Terahertz technology for military applications," *J. Mil. Inf. Sci.*, vol. 3, no. 1, pp. 1–15, 2015, doi: 10.17858/jmisci.58124.

[2] A. Y. Pawar, D. D. Sonawane, K. B. Erande, and D. V. Derle, "Terahertz technology and its applications," *Drug Invention Today*, vol. 5, no. 2, pp. 157–163, 2013, doi: 10.1016/j.dit.2013.03.009.

[3] E. Feiginov et al., *Semiconductor Terahertz Technology: Devices and Systems at Room Temperature Operation*. Hoboken, NJ, USA: Wiley, 2015.

[4] J. B. Gunn, "Microwave oscillations of current in III–V semiconductors," *Solid State Commun.*, vol. 1, no. 4, pp. 88–91, 1963, doi: 10.1016/0038-1098(63)90041-3.

[5] N. Priestley, K. Newsome, I. Dale, and P. Noton, "Gunn diode based surface mount 77 GHz oscillator for automotive applications," in *Proc. IEEE MTT-S Int. Microw. Symp. Dig.*, Jun. 2002, pp. 1–4, doi: 10.1109/MWSYM.2002.1012226.

[6] S. Chowdhury and U. K. Mishra, "Lateral and vertical transistors using the AlGaN/GaN heterostructure," *IEEE Trans. Electron Devices*, vol. 60, no. 10, pp. 3060–3066, Oct. 2013, doi: 10.1109/TED.2013.2277893.

[7] Y. Zhang et al., "Electrothermal simulation and thermal performance study of GaN vertical and lateral power transistors," *IEEE Trans. Electron Devices*, vol. 60, no. 7, pp. 2224–2230, Jul. 2013, doi: 10.1109/TED.2013.2261072.

[8] O. Yilmazoglu, K. Mutamba, D. Pavlidis, and T. Karaduman, "First observation of bias oscillations in GaN Gunn diodes on GaN substrate," *IEEE Trans. Electron Devices*, vol. 55, no. 6, pp. 1563–1567, Jun. 2008, doi: 10.1109/TED.2008.921253.

[9] K. Y. Xu, G. Wang, and A. M. Song, "Gunn oscillation in a self-switching nanodiode," *Appl. Phys. Lett.*, vol. 93, no. 23, 2008, Art. no. 233506.

[10] A. S. Hajo, O. Yilmazoglu, B. Samodi, A. Dadgar, F. Küppers, and T. Kussorow, "A new approach to achieve Gunn effect for GaN-based THz sources with high power," in *Proc. 44th Int. Conf. Infrared, Millimeter, THz Waves (IRMMW-THz)*, 2019, pp. 1–2, doi: 10.1109/IRMMW-THz.2019.8873720.

[11] W. Z. Lee, D. S. Ong, K. Y. Choo, O. Yilmazoglu, and H. L. Hartnagel, "Monte Carlo evaluation of GaN THz Gunn diodes," *IEEE Trans. Electron Devices*, vol. 68, pp. 1–6, 2021, doi: 10.1109/TED.2021.3086281.

[12] A. S. Hajo, D. Cicek, Y. Celik, A. Dadgar, O. Yilmazoglu, and S. Preu, "Nanostructured THz Gunn diode using integration of the side-contact and field-plate technologies," in *Proc. 48th Int. Conf. Infrared, Millimeter, Terahertz Waves (IRMMW-THz)*, 2023, pp. 1–2, doi: 10.1109/IRMMW-THz.2023.9318892.

[13] S. García-Sánchez et al., "Avoiding avalanche breakdown in planar GaN Gunn diodes by means of a substrate contact," *J. Phys. D, Appl. Phys.*, vol. 58, no. 1, 2024, Art. no. 15112, doi: 10.1088/1361-6463/ad809f.

[14] A. Khalid et al., "A planar Gunn diode operating above 100 GHz," *IEEE Electron Device Lett.*, vol. 28, no. 10, pp. 849–851, Oct. 2007, doi: 10.1109/LED.2007.904267.

[15] C. Li et al., "Enhancement of power and frequency in HEMT-like planar Gunn diodes by introducing extra delta-doping layers," *Microw. Opt. Technol. Lett.*, vol. 53, no. 7, pp. 1624–1626, Jul. 2011, doi: 10.1002/mop.26445.

[16] A. Khalid et al., "Terahertz oscillations in an In0.53Ga0.47As submicron planar Gunn diode," *J. Appl. Phys.*, vol. 115, no. 11, pp. 88–91, Mar. 2014, doi: 10.1063/1.4865095.

[17] V. N. Sokolov, K. W. Kim, V. A. Kochelap, and D. L. Woolard, "Terahertz generation in submicron GaN diodes within the limited space-charge accumulation regime," *J. Appl. Phys.*, vol. 98, no. 6, Sep. 2005, Art. no. 64507, doi: 10.1063/1.1995115.

[18] Y. Wang et al., "Ultra-short channel GaN high electron mobility transistor-like Gunn diode with composite contact," *J. Appl. Phys.*, vol. 116, no. 9, Sep. 2014, Art. no. 94502, doi: 10.1063/1.4892998.

[19] S. M. Sze, *Physics of Semiconductor Devices*, 2nd ed. Hoboken, NJ, USA: Wiley, 1981, doi: 10.1007/978-3-319-03002-9.

[20] A. Khalid et al., "$In_{0.53}Ga_{0.47}As$ planar Gunn diodes operating at a fundamental frequency of 164 GHz," *IEEE Electron Device Lett.*, vol. 34, no. 1, pp. 39–41, Jan. 2013.

[21] A. Khalid et al., "Terahertz oscillations in an $In_{0.53}Ga_{0.47}As$ submicron planar Gunn diode," *J. Appl. Phys.*, vol. 115, no. 11, 2014, Art. no. 114502.

[22] Y. Wang, L. A. Yang, W. Mao, S. Long, and Y. Hao, "Modulation of multidomain in AlGaN/GaN HEMT-like planar Gunn diode," *IEEE Trans. Electron Devices*, vol. 60, no. 5, pp. 1600–1606, May 2013, doi: 10.1109/TED.2013.2246092.

[23] Y. Wang, L. A. Yang, Z. Wang, J. Ao, and Y. Hao, "The modulation of multi-domain and harmonic wave in a GaN planar Gunn diode by recess layer," *Semicond. Sci. Technol.*, vol. 31, no. 2, 2016, Art. no. 25001, doi: 10.1088/0268-1242/31/2/025001.

[24] Y. Wang, L.-A. Yang, Z. Wang, J.-P. Ao, and Y. Hao, "The enhancement of the output characteristics in the GaN based multiple-channel planar Gunn diode," *Phys. Status Solidi A*, vol. 213, no. 5, pp. 1252–1258, 2015, doi: 10.1002/pssa.201532858.

[25] B. Li, H. Liu, Y. Alimi, and A. Song, "Simulation investigation of multiple-domain observed in $In_{0.23}Ga_{0.77}As$ planar Gunn diode," *Int. J. Hydrogen Energy*, vol. 41, no. 35, pp. 15772–15776, Jul. 2016, doi: 10.1016/j.ijhydene.2016.04.240.

[26] A. Khalid, C. Li, G. Dunn, N. Pilgrim, and D. S. R. Cumming, "Observation of multiple domains in a planar Gunn diode," in *Proc. Eur. Microw. Integr. Circuits Conf. (EuMIC)*, Sep. 2009, pp. 298–300, doi: 10.1109/EuMIC.2009.5338664.

[27] N. Balkan, M. C. Arikan, S. Gokden, V. Tilak, B. Schaff, and R. J. Shealy, "Energy and momentum relaxation of hot electrons in GaN/AlGaN," *J. Phys., Condens. Matter*, vol. 14, no. 13, pp. 3457–3468, 2002, doi: 10.1088/0953-8984/14/13/305.

[28] N. Balkan, M. C. Arikan, S. Gokden, V. Tilak, B. Schaff, and R. J. Shealy, "Energy and momentum relaxation of hot electrons in GaN/AlGaN," *J. Phys. Condens. Matter*, vol. 14, no. 13, pp. 3457–3464, 2002, doi: 10.1088/0953-8984/14/16/313.

[29] A. Matulionis and J. Liberis, "Microwave noise in AlGaN/GaN channels," *IEE Proc. Circuits Devices Syst.*, vol. 151, no. 1, pp. 148–154, Feb. 2004, doi: 10.1049/ip-cds:20045026.

[30] A. Matulionis and H. Morkoç, "Hot phonons in InAlN/AlN/GaN heterostructure 2DEG channels," in *Proc. SPIE OPTO*, 2009, pp. 1–8, doi: 10.1117/12.8204509.

[31] S. García, I. Íñiguez-de-la-Torre, S. Pérez, J. Mateos, and T. González, "Numerical study of sub-millimeter Gunn oscillations in InP and GaN vertical diodes: Dependence on DC bias, doping, and length," *J. Appl. Phys.*, vol. 114, no. 7, pp. 1–9, Oct. 2013, doi: 10.1063/1.4821865.

[32] G. Dunn and M. J. Kearney, "A theoretical study of differing active region doping profiles for W-band (75-110 GHz) InP Gunn diodes," *Semicond. Sci. Technol.*, vol. 18, no. 8, pp. 794–802, Aug. 2003, doi: 10.1088/0268-1242/18/8/304.

[33] C. Sevik and C. Bulutay, "Gunn oscillations in GaN channels," *Semicond. Sci. Technol.*, vol. 19, no. 4, p. 188, 2004, doi: 10.1088/0268-1242/19/4/306.

[34] I. Vurgaftman and J. R. Meyer, *Nitride Semiconductor Devices: Principles and Simulation*, J. Piprek, Eds., Weinheim, Germany: Wiley, 2007, pp. 49–67.
[35] L. Zhang, H. Jin, W. Yang, Z. Xie, H. Miao, and L. An, "Optical properties of single-crystalline α-Si3N4 nanobelts," *Appl. Phys. Lett.*, vol. 86, no. 6, pp. 1–3, Feb. 2005, doi: 10.1063/1.1850267.
[36] T. Inoue et al., "Polarization engineering on buffer layer in GaN-based heterojunction FETs," *IEEE Trans. Electron Devices*, vol. 55, no. 2, pp. 483–488, Feb. 2008, doi: 10.1109/TED.2008.915567.
[37] Y. Wang, L.-A. Li, C. Li, J.-P. Ao, X. Wang, and Y. Hao, "Thermal analysis of AlGaN/GaN hetero-structural Gunn diodes on different substrates through numerical simulation," *IEEE J. Electron Devices Soc.*, vol. 8, pp. 134–139, 2020.
[38] S. Arulkumaran et al., "Demonstration of submicron-gate AlGaN-GaN high-electron-mobility transistors on silicon with complementary metal–oxide–semiconductor-compatible non-gold metal stack," *Appl. Phys. Exp.*, vol. 6, no. 1, 2013, Art. no. 16501.
[39] V. Kumar et al., "AlGaN-GaN HEMTs on SiC with fT of over 120 GHz," *IEEE Electron Device Lett.*, vol. 23, no. 8, pp. 455–457, Aug. 2002.
[40] V. Kumar, W. Lu, F. A. Khan, R. Schwindt, E. Piner, and I. Adesida, "Recessed 0.25 μ-m gate AlGaN-GaN HEMTs on SiC with high gate-drain breakdown voltage using ICP-RIE," *Electron. Lett.*, vol. 37, no. 24, pp. 1483–1485, 2001.
[41] P. Cui et al., "Effect of different gate lengths on polarization coulomb field scattering potential in AlGaN/GaN heterostructure field-effect transistors," *Sci. Rep.*, vol. 8, p. 9036, Jun. 2018, doi: 10.1038/s41598-018-23354-4.
[42] A. dos Anjos, A. A. Pabon, and W. Van Noije, "2.45GHz low phase noise LC VCO design using flip chip on low-cost CMOS technology," in *Proc. IEEE 3rd Latin Amer. Symp. Circuits Syst. (LASCAS)*, Nov. 2012, pp. 1–4, doi: 10.1109/LASCAS.2012.6409291.
[43] M. Nick and A. Mortazawi, "Low phase-noise planar oscillators based on low-noise active resonators," *IEEE Trans. Microw. Theory Tech.*, vol. 58, no. 5, pp. 1133–1139, May 2010, doi: 10.1109/TMTT.2010.2045572.